Year 8 Mathematics – Yearly Exam Skills Checklist

Equations, Investigating Data and Mixed Topics

Name: _____ Teacher: _____

Tick one box for each skill: **Confident** – I can do this on my own. **Getting there** – I can do this with some help. **Need help** – I need to revise this.

Section A – Equations

#	Skill	Exam Q	Confident	Getting there	Need help
1	Solve one-step equations using inverse operations (+, −, ×, ÷)	Q1	☐	☐	☐
2	Solve two-step equations, including ones with division	Q2	☐	☐	☐
3	Solve equations that give a negative solution	Q3	☐	☐	☐
4	Check a solution by substituting it back into the equation	Q4	☐	☐	☐
5	Turn a worded “think of a number” problem into an equation	Q5a	☐	☐	☐
6	Write an equation for a real-life situation using a pronumeral	Q6a	☐	☐	☐
7	Solve equations made from worded problems and interpret the answer	Q5b, Q6b	☐	☐	☐

Section B – Investigating Data

#	Skill	Exam Q	Confident	Getting there	Need help
8	Read and interpret column graphs, including finding totals	Q7	☐	☐	☐
9	Read values from a line graph and work out changes between points	Q8	☐	☐	☐
10	Calculate amounts from percentages in a sector (pie) graph	Q9a	☐	☐	☐
11	Compare categories in a sector graph	Q9b	☐	☐	☐
12	Read a stem-and-leaf plot using its key	Q10a, c	☐	☐	☐
13	Find the median from a stem-and-leaf plot	Q10b	☐	☐	☐
14	Read a dot plot and find the mode	Q11	☐	☐	☐
15	Interpret “at least” and “more than” in data questions	Q10c, Q11b, Q12c	☐	☐	☐
16	Read a frequency table, find totals and the mode	Q12	☐	☐	☐
17	Calculate the mean of a data set	Q13a, Q14b	☐	☐	☐
18	Find the median of a data set with an even number of values	Q13b	☐	☐	☐
19	Find the mode and range of a data set	Q13c, d	☐	☐	☐
20	Identify an outlier and understand its effect on the mean	Q14	☐	☐	☐

Section C – Mixed Topics

#	Skill	Exam Q	Confident	Getting there	Need help
21	Use Pythagoras' theorem to find the hypotenuse	Q15a	☐	☐	☐
22	Use Pythagoras' theorem to test whether a triangle is right-angled	Q15b	☐	☐	☐
23	Multiply and divide decimals	Q16	☐	☐	☐
24	Evaluate powers and square roots	Q17	☐	☐	☐
25	Identify surds	Q18a	☐	☐	☐
26	Simplify surds using square factors, e.g. $\sqrt{50} = 5\sqrt{2}$	Q18b	☐	☐	☐
27	Simplify expressions by collecting like terms	Q19a	☐	☐	☐
28	Expand brackets using the distributive law	Q19b	☐	☐	☐
29	Substitute values into algebraic expressions, including squared terms	Q19c	☐	☐	☐
30	Find the area of a trapezium using the formula	Q20a	☐	☐	☐
31	Find the volume of a rectangular prism	Q20b	☐	☐	☐
32	Complete a table of values for a linear rule, including negative x-values	Q21	☐	☐	☐
33	Simplify ratios	Q22a	☐	☐	☐
34	Calculate average speed as a rate	Q22b	☐	☐	☐